Contact Information

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Christian Chávez

MSc student at Université de Sherbrooke, Québec, Canada
Full name: Kevin Christian Chávez Cadena
Citizenship: Ecuadorian

Curriculum Vitæ

Updated: October 7, 2026

Research Interests

Representation theory, category theory, algebraic topology, categorical and bicategorical approaches to topological data analysis.

Education

Master's degree in Mathematics, *Université de Sherbrooke*, Québec, Canada. **2025 – Present**
Research supervised by Emily Cliff and Thomas Brüstle.
Graduate academic record: Transcript.

Bachelor of Science in Mathematics, *Yachay Tech University*, Urcuquí, Ecuador. Graduated Summa Cum Laude. **2020 – 2025**
Undergraduate thesis: Homotopy Theory for Finite Categories.
Undergraduate academic record: Transcript & General summary.

Languages

Spanish (native), English (fluent), French (advanced)

Presentations

Math Club, Des points, des droites et un ballon : le théorème de Sylvester–Gallai, Université de Sherbrooke, Canada. Announcement. Slides. **29/09/2026**

SAG Seminar, Distance d'entrelacement et modules d'intervalle, Université de Sherbrooke, Canada. Announcement. **17/02/2026**

Undergraduate thesis defense, Homotopy Theory for Finite Categories, Yachay Tech University, Urcuquí, Ecuador. **04/04/2025**

VI Jornadas Ecuatorianas de Matemáticas, *Speaker: « Homotopy Theory for Finite Categories »*, Escuela Superior Politécnica de Chimborazo, Ecuador. Acceptance letter. **28/11/2024**

Conference talk, *Applications of Topological Data Analysis: Exploring Persistent Homology in Social Inequality and Neural Networks Optimization*, Second International Conference on Applied and Computational Mathematics for Students (ICACMS), Universidad Politécnica Estatal del Carchi, Ecuador. Certificate. **30/05/2024**

Seminar talk, *Cryptography: The Art of Hiding Messages*, Seminar on Introduction to Number Theory, Math Club, Yachay Tech University. Certificate. **31/07/2023**

Scientific poster, *The Cycloid: A Bit of History and Some of its Particularities*, Open house "Todos Nos Sumamos", Yachay Tech University. Poster. Certificate. **09/03/2023**

Awards & Fellowships

ISM Scholarship, *Awarded to top graduate students in the mathematical sciences in Québec (total: 7 000 CAD, including 3 500 CAD funded by the ISM)*, Institut des sciences mathématiques (ISM), Award letter. **2026 – 2027**

ISM Travel Grant, *500 CAD to attend the workshop* Computable Invariants of Quiver Representations*, Lorentz Center, Leiden, The Netherlands*, Institut des sciences mathématiques (ISM), Award letter. **2026**

Valorisation des travaux étudiants, *625 CAD Faculty of Science award to present research at the workshop* Computable Invariants of Quiver Representations*, Lorentz Center, Leiden, The Netherlands*, Faculty of Science, Université de Sherbrooke, Award letter. **2026**

Valedictorian, *Best GPA of the 2020–2025 cohort*, Yachay Tech University. **2025**

Graduated Summa Cum Laude, Yachay Tech University. **2025**

Academic Excellence Scholarship, *Awarded for 9 consecutive semesters (total: 8 100 USD)*, Yachay Tech University, Certificate. **2020 – 2024**

Conferences & Workshops Attended

Workshop: Computable Invariants of Quiver Representations, *Lorentz Center*, Leiden, The Netherlands. Workshop webpage. Poster presentation. **07/2026**

XXVIII^{e} Colloque panquébécois de l'ISM, *Université du Québec à Trois-Rivières*, Québec, Canada. **05/2026**

35th Meeting on the Representation Theory of Algebras and Related Topics, *Université de Sherbrooke*, Québec, Canada. **10/2025**

XIX SEDEM Mathematics Olympiad, *Member of the Organizing Committee*, Yachay Tech University. Certificate. **06/2024**

V Jornadas Ecuatorianas de Matemáticas, *Member of the Organizing Committee*, Yachay Tech University. Certificate. **11/2023**

XVIII SEDEM Mathematics Olympiad, *Trainer*, Yachay Tech University. Certificate. **05/2023**

First Ecuadorian Congress of Young Researchers (CEJI22), *Attendee*, Office of the Vice-Rector for Research and Innovation & IEEE EMBS Yachay Tech. Certificate. **03/2022**

International Youth Math Challenge 2021, *International mathematics competition*, Certificate. **10/2021**

Research Experience

Research Assistant, *Research project "Examining Inequality in Ecuador: An Application of Topological Data Analysis".* Director: Pablo Rosero, M.Sc., Yachay Tech University. **2023 – 2025**

Teaching Experience

Discrete Mathematics, *Teaching Assistant for MAT120*, Université de Sherbrooke. **2026 (Fall)**

Linear Algebra, *Grading Assistant for MAT253*, Université de Sherbrooke. **2026 (Winter)**

Linear Algebra, *Grading Assistant for MAT193*, Université de Sherbrooke. **2025 (Fall)**

Abstract Algebra, *Teaching Assistant*, Yachay Tech University, Ecuador. **2024 (S2)**

Abstract Algebra, *Teaching Assistant*, Yachay Tech University, Ecuador. **2024 (S1)**

General Topology, *Teaching Assistant*, Yachay Tech University, Ecuador. **2023 (S2)**

Linear Algebra, *Teaching Assistant*, Yachay Tech University, Ecuador. **2023 (S1)**

Lecture Notes

Abstract Algebra, *Lecture notes written with Professor Pablo Rosero*, Repository. **2024 (S2)**

Linear Algebra, *"Álgebra Lineal y Problemas Resueltos"*, Kevin Christian Chávez & Johana Romero-Leiton, Editorial Universidad de Nariño, UDENAR, Colombia, Lecture notes written with the instructor. Publication. **2022**

Extracurricular Activities

Participant in the REL-AL project, *Interactive linear algebra textbook. Contribution: figures and GeoGebra applets. Website*. **01/2024 – 01/2025**

President of the Yachay Tech Math Club, *Yachay Tech University. Club website*. **01/2024 – 01/2025**

Instructor, *Introductory course « Point-Set Topology »*, Math Club, Course webpage, Yachay Tech University. **01/2025**

Instructor, *Mini-course « Writing Scientific Articles with LaTeX »*, Math Club, Course webpage, Yachay Tech University. **10/2024**

Instructor, *Mini-course « Writing Scientific Articles with LaTeX »*, Math Club, Yachay Tech University. **03/2024**

Volunteer, outreach project "Matemáticas Recreativas", *Agreement between Yachay Tech University and Unidad Educativa Teodoro Gómez de la Torre*, Ibarra, Ecuador. Director: Rodolfo Gallo, Ph.D. Certificate. **2022 – 2023**

Tutor for the Clavemat community, *Professional practice; tutoring undergraduates in calculus and linear algebra*, Escuela Politécnica Nacional, Quito, Ecuador. Certificate. **2021**

Extracurricular Courses

Commutative Algebra: Toward Algebraic Geometry, *Physics Latam*, Course webpage. Admission letter. **09/2024 – 01/2025**

Data Scientist with Python, *Data manipulation, visualization, and analysis with Python*, DataCamp. Certificate. **2023**

Matlab Onramp, *Self-paced training course*, MathWorks Training Services. Certificate. **2021**

Data Analysis, *Importance of data analysis; tools, formats, and best practices for using information*, AIESEC, Ecuador. Certificate. **2021**

Software and Technical Skills

LEAN 4, (La/Xe)TeX, Linux, Git(Hub), (Neo)Vim, Bash, Matlab, Python, C, R

Inkscape, PreTeXt